



#1

Antenna

horn and pyramidal horn antennas

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E-Plane and H-Plane Radiation Patterns

The radiation pattern plots illustrate the field strength (usually in dBi or dB normalized to the maximum) in two principal planes: the E-plane and the H-plane.

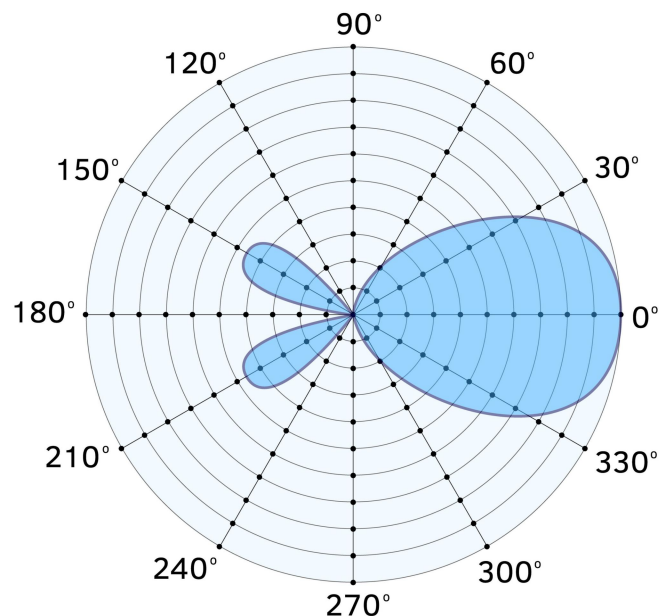
1. Defining the Planes

E-Plane (Electric Field Plane): The plane containing the Electric field vector and the direction of maximum radiation. This plane is parallel to the wider flare dimension in a typical pyramidal horn.

H-Plane (Magnetic Field Plane): The plane containing the Magnetic field vector and the direction of maximum radiation. This plane is perpendicular to the E-plane and parallel to the narrower flare dimension.

2. General Plot Characteristics

The plots are typically shown on a polar coordinate system (angle vs. amplitude in dB) or sometimes a rectangular plot.



Main Lobe: A dominant, narrow beam centered in the direction of maximum radiation (0° on the plot).

Side Lobes: Smaller lobes appearing away from the main beam. Horn antennas are designed to minimize these.

Beamwidth: The angular width of the main lobe, usually measured at the Half-Power Beamwidth (HPBW), which is the angle between the points where the power drops to half the maximum value (-3dB).

Pyramidal Horn Antenna Section

The pyramidal horn is a rectangular waveguide that is flared (widened) in both the E-plane and the H-plane directions.

Structure

Flare: Flared on all four sides, resulting in a four-sided pyramidal shape.

Cross-Section: The rectangular cross-section of the waveguide expands uniformly in both the broader dimension (a) and the narrower dimension (b) as it progresses toward the aperture.

- a is AH (H-plane aperture width)

- B is AE (E-plane aperture height)

Aperture: The final opening is a large rectangle.

Radiation Pattern: Provides a narrow, symmetrical beam (or nearly symmetrical) in both the E-plane and H-plane, leading to high directivity and high gain. It's often used as a standard gain antenna for calibration.

Matlab code :

```
%% Pyramidal Horn Antenna Radiation Patterns
% Simplified code with single aperture value and aperture comparison
clear all; close all; clc;

%% Antenna Parameters (Single Value)
lambda = 1;           % Wavelength (normalized to 1)
a_wg = 0.5;           % Waveguide width in wavelengths
b_wg = 0.25;          % Waveguide height in wavelengths
a_horn = 3.0 * lambda; % Horn width (single value)
b_horn = 2.0 * lambda; % Horn height (single value)
eta_a = 0.5;          % Aperture efficiency
efficiency = 0.7;      % Overall antenna efficiency

%% Angle Definitions
theta = linspace(-pi/2, pi/2, 361); % Elevation angle (radians)
theta_deg = theta * 180/pi;         % Convert to degrees
k = 2*pi/lambda;

%% Calculate Patterns (Single Aperture)
E_h = zeros(size(theta));
E_e = zeros(size(theta));
% H-plane pattern ( $\phi = 0^\circ$ )
for i = 1:length(theta)
    u = k*a_horn/2 * sin(theta(i));
    if abs(u) < 1e-10
        E_h(i) = 1;
    else
        E_h(i) = abs(sinc(u/pi) * cos(theta(i)));
    end
end
% E-plane pattern ( $\phi = 90^\circ$ )
for i = 1:length(theta)
    v = k*b_horn/2 * sin(theta(i));
    if abs(v) < 1e-10
        E_e(i) = 1;
    else
        E_e(i) = abs(sinc(v/pi) * cos(theta(i)));
    end
end
% Normalize patterns
E_h = E_h / max(E_h);
```

```
E_e = E_e max(E_e);
```

```
%% Calculate Directivity and Gain (Single Aperture)
```

```
A_phys = a_horn b_horn; % Physical aperture area
```

```
A_eff = A_phys eta_a; % Effective aperture area
```

```
D_max = 4 pi A_eff lambda^2; % Maximum directivity
```

```
D_max_dBi = 10 log10(D_max); % Directivity in dBi
```

```
G_max = D_max efficiency; % Maximum gain
```

```
G_max_dBi = 10 log10(G_max); % Gain in dBi
```

```
%% Plot 1 E-plane and H-plane Patterns (Single Aperture)
```

```
figure;
```

```
% Rectangular plot
```

```
plot(theta_deg, E_e, 'b-', 'LineWidth', 2);
```

```
hold on;
```

```
plot(theta_deg, E_h, 'r-', 'LineWidth', 2);
```

```
xlabel('Angle  $\theta$  (degrees)', 'FontSize', 12);
```

```
ylabel('Normalized Field Pattern', 'FontSize', 12);
```

```
title(sprintf('Pyramidal Horn %.1f $\lambda \times$  %.1f $\lambda$ ', a_hornlambda, b_hornlambda), 'FontSize', 14);
```

```
legend('E-plane ( $\phi = 90^\circ$ )', 'H-plane ( $\phi = 0^\circ$ )', 'Location', 'best');
```

```
grid on;
```

```
xlim([-90, 90]);
```

```
ylim([0, 1]);
```

```
set(gca, 'FontSize', 11);
```

```
% Polar plot
```

```
figure;
```

```
polarplot(theta, E_e, 'b', 'LineWidth', 2);
```

```
hold on;
```

```
polarplot(theta, E_h, 'r', 'LineWidth', 2);
```

```
title(sprintf('Radiation Patterns %.1f $\lambda \times$  %.1f $\lambda$ ', a_hornlambda, b_hornlambda), 'FontSize', 14);
```

```
legend('E-plane', 'H-plane', 'Location', 'best');
```

```
rlim([0, 1]);
```

```
set(gca, 'FontSize', 11);
```

```
%% Plot 2 Directivity and Gain (Single Aperture)
```

```
figure;
```

```
% Create a bar plot
```

```
metrics = {'Max Directivity', 'Antenna Gain'};
```

```
values_dB = [D_max_dBi, G_max_dBi];
```

```
values_linear = [D_max, G_max];
```

```
subplot(1,2,1);
```

```

bar(values_dB, 'FaceColor', [0.2 0.6 0.8]);
ylabel('Value (dBi)', 'FontSize', 12);
title('Directivity and Gain (dB Scale)', 'FontSize', 14);
set(gca, 'XTickLabel', metrics);
grid on;
% Add value labels
for i = 1:length(values_dB)
    text(i, values_dB(i) + 0.5, sprintf('%.2f dBi', values_dB(i)), ...
        'HorizontalAlignment', 'center', 'FontSize', 11, 'FontWeight', 'bold');
end
subplot(1,2,2);
bar(values_linear, 'FaceColor', [0.8 0.4 0.2]);
ylabel('Value (Linear)', 'FontSize', 12);
title('Directivity and Gain (Linear Scale)', 'FontSize', 14);
set(gca, 'XTickLabel', metrics);
grid on;
% Add value labels
for i = 1:length(values_linear)
    text(i, values_linear(i) + 0.5, sprintf('%.2f', values_linear(i)), ...
        'HorizontalAlignment', 'center', 'FontSize', 11, 'FontWeight', 'bold');
end
%% Normalized Directivity Patterns for Different Aperture Sizes
figure;
% Different horn aperture sizes to compare (H-plane width variation)
a_sizes = [5.5, 4.5, 3.5]; %  $\lambda$  units (width only, height varies with ratio)
labels = {'a = 5.5 $\lambda$ ', 'a = 4.5 $\lambda$ ', 'a = 3.5 $\lambda$ '};
colors = {'r', 'g', 'b'};
line_styles = {'-', '--', ''};
% Fixed height-to-width ratio for pyramidal horn
height_ratio = 0.667; %  $b/a$  ratio
% Calculate normalized directivity patterns for each aperture size
theta_norm = linspace(0, 30, 200); % Only positive angles, 0 to 30 degrees
D_norm_all = zeros(length(a_sizes), length(theta_norm));
for size_idx = 1:length(a_sizes)
    a_current = a_sizes(size_idx) * lambda;
    b_current = a_current * height_ratio;
    % Calculate H-plane pattern for this aperture size
    E_h_current = zeros(size(theta_norm));
    for i = 1:length(theta_norm)

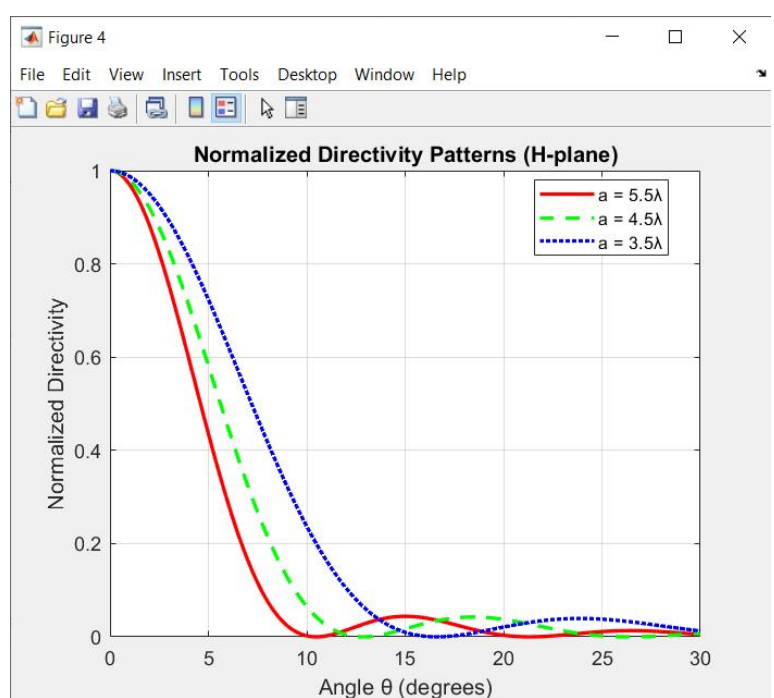
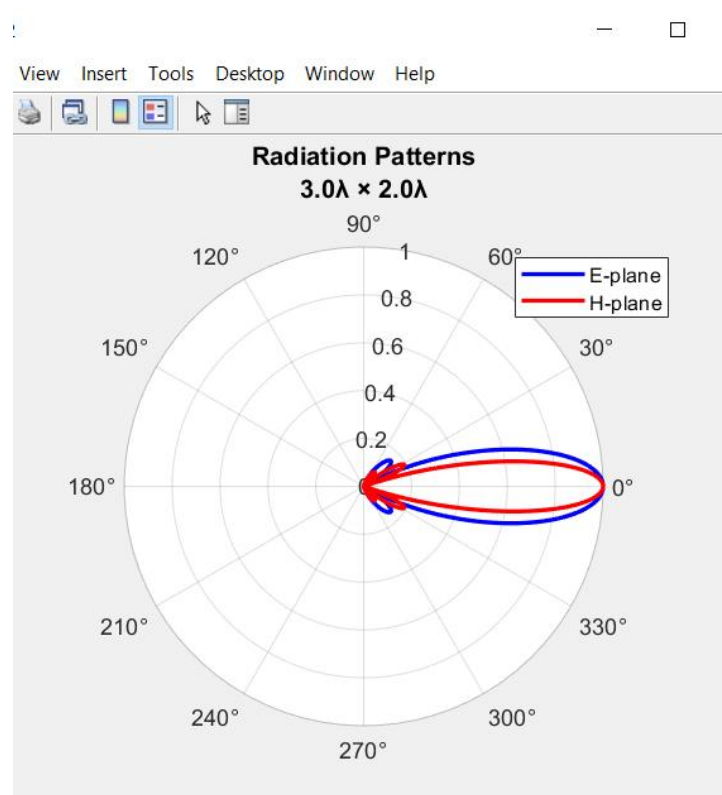
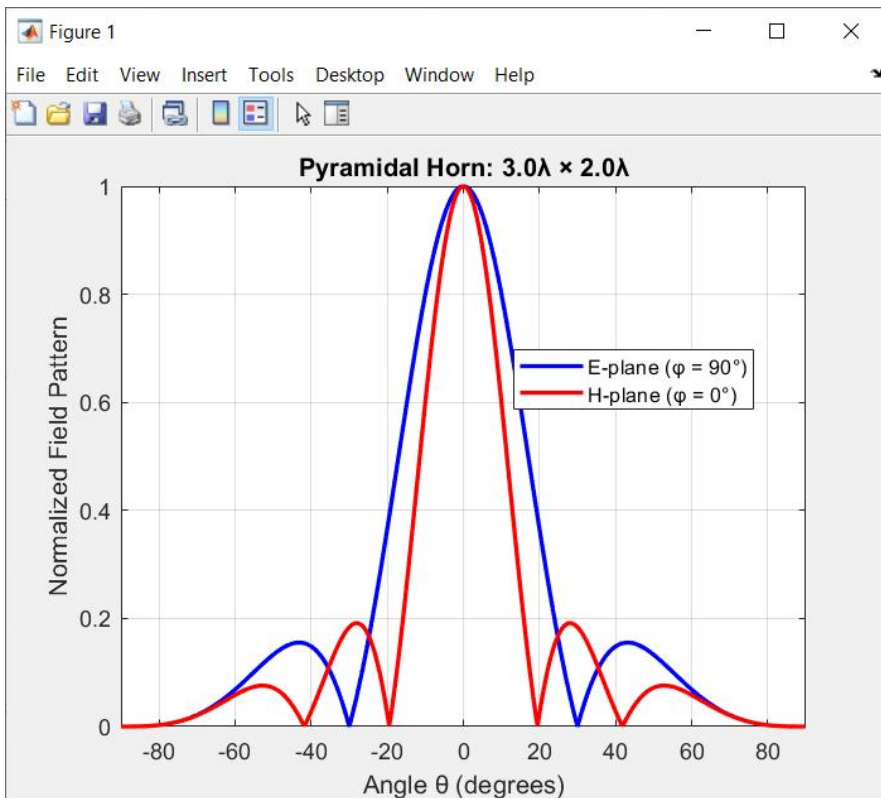
```

```

theta_rad = deg2rad(theta_norm(i));
u = ka_current2 * sin(theta_rad);
if abs(u) > 1e-10
    E_h_current(i) = 1;
else
    E_h_current(i) = abs(sinc(u*pi) * cos(theta_rad));
end
end
% Normalize the pattern
E_h_current = E_h_current / max(E_h_current);
% Calculate normalized directivity pattern ( $E^2$ )
D_norm_all(size_idx, :) = E_h_current.^2;
end
% Plot normalized directivity patterns
for size_idx = 1:length(a_sizes)
    plot(theta_norm, D_norm_all(size_idx, :), ...
        [colors{size_idx} line_styles{size_idx}], ...
        'LineWidth', 2, 'DisplayName', labels{size_idx});
    hold on;
end
xlabel('Angle  $\theta$  (degrees)', 'FontSize', 12);
ylabel('Normalized Directivity', 'FontSize', 12);
title('Normalized Directivity Patterns (H-plane)', 'FontSize', 14);
legend('Location', 'best');
grid on;
xlim([0, 30]);
ylim([0, 1]);
set(gca, 'FontSize', 11);
%% Display Only Necessary Results
fprintf('Pyramidal Horn Antennan');
fprintf('=====\n');
fprintf('Aperture %.1f $\lambda$   $\times$  %.1f $\lambda$ n', a_hornlambda, b_hornlambda);
fprintf('Directivity %.2f (%.2f dBi)n', D_max, D_max_dBi);
fprintf('Gain %.2f (%.2f dBi)n', G_max, G_max_dBi);
fprintf('n');
fprintf('Directivity Comparison (H-plane)n');
fprintf(' a = 5.5 $\lambda$  Pattern widensn');
fprintf(' a = 4.5 $\lambda$  Intermediaten');
fprintf(' a = 3.5 $\lambda$  Pattern narrowsn');

```

Results :



Command Window

Pyramidal Horn Antenna

=====

Aperture: $3.0\lambda \times 2.0\lambda$

Directivity: 37.70 (15.76 dBi)

Gain: 26.39 (14.21 dBi)

Directivity Comparison (H-plane):

$a = 5.5\lambda$: Pattern widens

$a = 4.5\lambda$: Intermediate

$a = 3.5\lambda$: Pattern narrows

Sectoral Horn Antenna Section

A sectoral horn is a type of horn that is flared in only one dimension while the other dimension remains constant (parallel walls). This creates an asymmetric beam shape.

There are two subtypes of sectoral horns, named after the plane in which they flare:

1. E-Plane Sectoral Horn

Flare: Only flared in the direction of the Electric field vector (the height, or the narrower dimension b of the rectangular waveguide).

Non-Flared: The walls in the H-plane (the width, or the broader dimension a) remain parallel.

Beam Shape: The beam is narrow in the E-plane (flared dimension) but wide in the H-plane (non-flared dimension). This results in a fan-shaped beam.

2. H-Plane Sectoral Horn

Flare: Only flared in the direction of the Magnetic field vector (the width, or the broader dimension a of the rectangular waveguide).

Non-Flared: The walls in the E-plane (the height, or the narrower dimension b) remain parallel.

Beam Shape: The beam is narrow in the H-plane (flared dimension) but wide in the E-plane (non-flared dimension). This also results in a fan-shaped beam.

Matlab code :

```
%% H-plane Sectoral Horn Antenna Radiation Patterns
% Simplified code with rectangular and polar plots only
clear all; close all; clc;

%% Antenna Parameters
lambda = 1;      % Wavelength (normalized to 1)
a1 = 0.5;        % Waveguide width in wavelengths
b = 0.25;        % Waveguide height in wavelengths
a = 2*lambda;    % Horn aperture width
k = 2*pi/lambda; % Wave number
efficiency = 0.7; % Antenna efficiency (70% typical)

%% Angle Definitions
theta = linspace(-pi/2, pi/2, 361); % Elevation angle (radians)
theta_deg = theta * 180/pi;         % Convert to degrees

%% H-plane (phi=0°) Pattern Calculation
```

```

E_h = zeros(size(theta));
for i = 1:length(theta)
    u = k*a/2 * sin(theta(i));
    if abs(u) < 1e-10
        E_h(i) = 1; % Avoid division by zero
    else
        E_h(i) = abs(sinc(u/pi) * cos(theta(i)));
    end
end

%% E-plane (phi=90°) Pattern Calculation
E_e = zeros(size(theta))
for i = 1:length(theta)
    v = k*b/2 * sin(theta(i));
    if abs(v) < 1e-10
        E_e(i) = 1;
    else
        E_e(i) = abs(sinc(v/pi) * cos(theta(i)));
    end
end

%% Normalize patterns
E_h = E_h / max(E_h);
E_e = E_e / max(E_e);

%% Calculate Directivity - Practical Method
% Using aperture efficiency method (most practical for horn antennas)
A_phys = a * b; % Physical aperture area ( $\lambda^2$ )
eta_a = 0.5; % Aperture efficiency (typical for horns)
A_eff = A_phys * eta_a; % Effective aperture area ( $\lambda^2$ )
% Directivity from effective aperture
D_max = 4 * pi * A_eff / lambda^2; % Maximum directivity
D_max_dBi = 10 * log10(D_max); % Maximum directivity in dBi
% Gain calculation
G_max = D_max * efficiency; % Maximum gain
G_max_dBi = 10 * log10(G_max); % Maximum gain in dBi

%% Rectangular Plot (Linear Scale)
figure;
plot(theta_deg, E_e, 'b-', 'LineWidth', 2);
hold on;
plot(theta_deg, E_h, 'r-', 'LineWidth', 2);
xlabel('Angle  $\theta$  (degrees)', 'FontSize', 12);

```

```

ylabel('Normalized Field Pattern', 'FontSize', 12);
title('E-plane and H-plane Radiation Patterns', 'FontSize', 14);
legend('E-plane ( $\phi = 90^\circ$ )', 'H-plane ( $\phi = 0^\circ$ )', 'Location', 'best');
grid on;
xlim([-90, 90]);
ylim([0, 1]);
set(gca, 'FontSize', 11);
%% Polar Plots
figure;
% E-plane & H-plane polar plot
polarplot(theta, E_e, 'b', 'LineWidth', 2);
hold on;
polarplot(theta, E_h, 'r', 'LineWidth', 2);
title('E-plane ( $\phi = 90^\circ$ ) and H-plane Pattern ( $\phi = 0^\circ$ )', 'FontSize', 14);
legend('E-plane ( $\phi = 90^\circ$ )', 'H-plane ( $\phi = 0^\circ$ )', 'Location', 'best');
rlim([0, 1]);
set(gca, 'FontSize', 11);
%% Calculate and Display Beamwidths
% Find 3dB points (0.707 of maximum)
idx_3dB_e = find(E_e >= 0.707);
if ~isempty(idx_3dB_e)
    theta_3dB_e = [theta_deg(idx_3dB_e(1)), theta_deg(idx_3dB_e(end))];
    HPBW_e = diff(theta_3dB_e);
else
    HPBW_e = NaN;
end
idx_3dB_h = find(E_h >= 0.707);
if ~isempty(idx_3dB_h)
    theta_3dB_h = [theta_deg(idx_3dB_h(1)), theta_deg(idx_3dB_h(end))];
    HPBW_h = diff(theta_3dB_h);
else
    HPBW_h = NaN;
end
%% Plot 4: Directivity Comparison Plot
figure;
% Create a bar plot for key metrics
metrics = {'Max Directivity', 'Antenna Gain'};
values_dB = [D_max_dBi, G_max_dBi];
values_linear = [D_max, G_max];

```

```

subplot(2,1,1);
bar(values_dB, 'FaceColor', [0.2 0.6 0.8]);
ylabel('Value (dBi)', 'FontSize', 12);
title('Directivity and Gain with Efficiency = 0.7 (dB Scale)', 'FontSize', 14);
set(gca, 'XTickLabel', metrics);
grid on;
% Add value labels on bars
for i = 1:length(values_dB)
    text(i, values_dB(i) + 0.5, sprintf('%.2f dBi', values_dB(i)), ...
        'HorizontalAlignment', 'center', 'FontSize', 11, 'FontWeight', 'bold');
end
subplot(2,1,2);
bar(values_linear, 'FaceColor', [0.8 0.4 0.2]);
ylabel('Value (Linear)', 'FontSize', 12);
title('Directivity and Gain with Efficiency = 0.7 (Linear Scale)', 'FontSize', 14);
set(gca, 'XTickLabel', metrics);
grid on;
% Add value labels on bars
for i = 1:length(values_linear)
    text(i, values_linear(i) + 0.5, sprintf('%.2f', values_linear(i)), ...
        'HorizontalAlignment', 'center', 'FontSize', 11, 'FontWeight', 'bold');
end
%% NEW PLOT: Normalized Directivity Patterns for Different Aperture Sizes
figure;
% Different horn aperture sizes to compare
a_sizes = [5.5, 4.5, 3.5]; %  $\lambda$  units
labels = {'a = 5.5 $\lambda$ ', 'a = 4.5 $\lambda$ ', 'a = 3.5 $\lambda$ '};
colors = {'r', 'g', 'b'};
line_styles = {'-', '--', ':'};
% Calculate normalized directivity patterns for each aperture size
theta_norm = linspace(0, 30, 200); % Only positive angles, 0 to 30 degrees
D_norm_all = zeros(length(a_sizes), length(theta_norm));
for size_idx = 1:length(a_sizes)
    a_current = a_sizes(size_idx) * lambda;
    % Calculate H-plane pattern for this aperture size
    E_h_current = zeros(size(theta_norm));
    for i = 1:length(theta_norm)
        theta_rad = deg2rad(theta_norm(i));
        u = k*a_current/2 * sin(theta_rad);

```

```

    if abs(u) < 1e-10
        E_h_current(i) = 1;
    else
        E_h_current(i) = abs(sinc(u/pi) * cos(theta_rad));
    end
end
% Normalize the pattern
E_h_current = E_h_current / max(E_h_current);
% Calculate normalized directivity pattern ( $|E|^2$ )
D_norm_all(size_idx, :) = E_h_current.^2;
end
% Plot normalized directivity patterns
for size_idx = 1:length(a_sizes)
    plot(theta_norm, D_norm_all(size_idx, :), ...
        [colors{size_idx} line_styles{size_idx}], ...
        'LineWidth', 2, 'DisplayName', labels{size_idx});
    hold on;
end
xlabel('Angle  $\theta$  (degrees)', 'FontSize', 12);
ylabel('Normalized Directivity', 'FontSize', 12);
title('Normalized Directivity Patterns', 'FontSize', 14);
legend('Location', 'best');
grid on;
xlim([0, 30]);
ylim([0, 1]);
set(gca, 'FontSize', 11);
%% Display Results
fprintf('H-plane Sectoral Horn Antenna\n');
fprintf('=====\n');
fprintf('Waveguide: %.2f $\lambda$   $\times$  %.2f $\lambda$ \n', a1, b);
fprintf('Horn aperture width: %.2f $\lambda$ \n', a/lambda);
fprintf('\nBeamwidths:\n');
fprintf('E-plane HPBW: %.2f $^\circ$ \n', HPBW_e);
fprintf('H-plane HPBW: %.2f $^\circ$ \n', HPBW_h);
fprintf('Max Directivity: %.2f\n', D_max);
fprintf('Max Directivity(dB): %.2f dBi\n', D_max_dBi);
fprintf('Max Gain: %.2f\n', G_max);
fprintf('Max Gain(dB): %.2f dBi\n', G_max_dBi);

```

Results :

Command Window

```
H-plane Sectoral Horn Antenna
=====
Waveguide: 0.50λ × 0.25λ
Horn aperture width: 2.00λ

Beamwidths:
E-plane HPBW: 84.00°
H-plane HPBW: 24.00°
Max Directivity: 3.14
Max Directivity(dBi): 4.97 dBi
Max Gain: 2.20
Max Gain(dBi): 3.42 dBi
```

